

# Validity Analysis: WXIMPACTBENCH

Target context: Gran Chaco Tri-Border Emergency Response Zone

Overall risk: **HIGH**

## Deployment Context

**Use case:** Use Case: Early identification of damage type caused by a disaster and reported in local news and social media posts from Northwest Argentina, Paraguay and South Bolivia (e.g. Gran Chaco regions).

**Target population:** Target population: National emergency responders, gendarmerie, civil forces, technical practitioners in public emergency sector from Argentina, Bolivia or Paraguay. They all would speak and write in Spanish, Andean variety, and would have an intermediate level of English.

## Dimension Scores

| Dimension                                                                                        | Score      | Rating           | Confidence |
|--------------------------------------------------------------------------------------------------|------------|------------------|------------|
| Input Ontology  | 1          | Serious concern  | high       |
| Input Content   | 1          | Serious concern  | high       |
| Input Form      | 1          | Serious concern  | high       |
| Output Ontology                                                                                  | 2          | Significant gaps | high       |
| Output Content  | 1          | Serious concern  | high       |
| Output Form   | 3          | Moderate gaps    | medium     |
| <b>Average</b>                                                                                   | <b>1.5</b> |                  |            |

 = highest concern     = strongest dimension

## Dimension Key

| Abbr. | Dimension       | Definition                                                                                                   |
|-------|-----------------|--------------------------------------------------------------------------------------------------------------|
| IO    | Input Ontology  | Whether the benchmark's test case categories cover the query types expected in deployment.                   |
| IC    | Input Content   | Whether individual datapoint content is culturally and contextually appropriate for the target region.       |
| IF    | Input Form      | Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.             |
| OO    | Output Ontology | Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries. |
| OC    | Output Content  | Whether ground-truth labels align with the judgments of regional stakeholders.                               |
| OF    | Output Form     | Whether the expected output modality matches regional deployment needs and accessibility.                    |

## Overall Summary

WXIMPACTBENCH was constructed exclusively from English-language Canadian historical newspaper content with annotation by Canadian academic climate-history researchers. The Gran Chaco Tri-Border Emergency Response deployment requires processing Rio Platense Spanish social media and local news with Guaraní/Quechua/Wichí lexical borrowing, applies consequence-framing-dependent label logic, prioritizes cross-border displacement and indigenous community access disruption, and designates AFE/SINAGIR national coordinators as authoritative labelers — none of which are represented in the benchmark. Five of six dimensions show critical or major mismatches; only the output form structure (multi-label

simultaneous binary classification) is reusable. Dataset analysis confirms label-boundary inconsistencies even within the Canadian context (explicit human casualties labeled all-zero in multiple cases), suggesting that the annotation philosophy itself does not match emergency-responder operational salience.

## Practical Guidance

### What This Benchmark Measures

WXIMPACTBENCH measures LLM ability to assign six broad weather-impact categories to formal English-language historical newspaper prose under a Canadian-academic annotation philosophy emphasizing 'explicit mentions' of direct physical effects. The strongest dimension for the deployment is Output Form (score 3): the multi-label simultaneous binary classification output structure aligns with the minimum acceptable deployment output, and dataset analysis confirms the schema is operationally usable. The benchmark also provides a documented six-category template that, with significant adaptation, could serve as a starting point for Gran Chaco taxonomy design.

### Construct Depth

Construct depth is shallow for the deployment context. The benchmark probes one slice of disaster-impact understanding — English formal-prose multi-label classification — without addressing input register diversity (Spanish social media), regional language phenomena (indigenous lexical borrowing), regional event types (locust outbreaks, Chaco fires, cross-border displacement), or annotator-population validity (AFE/SINAGIR coordinators absent). Even within its construct, the dataset reveals label-boundary inconsistencies [D6, D9, D16, D18] and missing IAA documentation that limit how confidently any score can be interpreted as a measure of impact-classification competence. The benchmark provides almost no transferable evidence about how a model would perform on the deployment's actual input distribution.

### What Else You Need

The benchmark cannot stand alone for this deployment. Required supplementation: (a) construct a Spanish-language Gran Chaco disaster social media evaluation set with AFE/SINAGIR-coordinator-validated labels (addresses IC, IF, OC); (b) extend or redesign the impact taxonomy to include cross-border displacement, indigenous-community-access disruption, locust outbreaks, and Chaco fires (addresses IO, OO); (c) define role-differentiated evaluation metrics distinguishing field-gendarme precision needs from national-coordinator aggregate recall (addresses OF); (d) evaluate Spanish social-media-tuned models (e.g., RoBERTuito [WEB-11]) alongside multilingual LLMs in the deployment register; (e) document inter-annotator agreement and consequence-framing decision rules for the regional taxonomy.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D6     | A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — Storm narrative with human casualty but annotated all-zero — potential label discrepancy                                                                                                                                                               |
| D9     | residents of southwestern Quebec have been plunged back into the deep freeze Overnight lows in Montreal were expected to drop to minus-16 Celsius last night and tumble as far as minus-30C on the South Shore, where more than one million people struggle on without power or heat — 1998 Quebec ice storm aftermath — one million without power/heat, annotated all-zero                                                                       |
| D16    | More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged                                                                       |
| D18    | A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — 1992 Mediterranean flash flood — political and ecological but not infrastructural ("swept away cars," "uprooted trees") — label boundary question |
| WEB-11 | RoBERTuito (500M+ Spanish tweets) is the most relevant existing model for the deployment's input register, but is not disaster-domain-specific and not regionally evaluated ( <a href="https://aclanthology.org/2022.lrec-1.785/">https://aclanthology.org/2022.lrec-1.785/</a> )                                                                                                                                                                 |

# Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Gran Chaco Tri-Border Emergency Response Zone

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark's six impact categories (Infrastructural, Political, Financial, Ecological, Agricultural, Human Health) and 15 LDA-derived weather event types were derived exclusively from Canadian newspaper content [Q11, Q101]. The Gran Chaco deployment requires categories that capture cross-border displacement along the Pilcomayo and Bermejo rivers, indigenous community access disruption, locust outbreaks, Chaco fires (incendios), and strategic economic route disruption — none of which are represented in the benchmark's taxonomy or sampled data. The deployment user explicitly identified these as primary disaster news categories, and the regional context confirms locust outbreaks and cross-border displacement are 'ABSENT from wximpactbench'. Dataset analysis confirms the Political category is operationalized via Canadian municipal exemplars (e.g., snow-removal budget debates [D10]) with no border-integrity analog.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                 |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q11  | "The articles are selected by topic modeling, including six impact categories (infrastructural, political, financial, ecological, agricultural, and human health), which are informed by previous studies (Imran et al., 2016a) and align with modern disaster impact assessment frameworks (Silva et al., 2022)." (p.2) |
| Q101 | "Using Latent Dirichlet Allocation, the dataset was categorized into 15 primary weather event types." (p.14)                                                                                                                                                                                                             |
| D10  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                    |

## Strengths

- Six broad impact categories nominally subsume flooding and fire damage descriptions, providing partial coverage of generic disaster consequences [Q11, Q25]
- Multi-label structure allows a single event to attract multiple labels simultaneously [Q28], which is structurally compatible with consequence-framing-dependent assignment even though the specific labels do not match
- Dataset confirms multi-label co-occurrence works in practice (e.g., infra+pol+fin assigned together [D10])

| ID  | Evidence                                                                                                                                                                                                                                                                                                                 |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q11 | "The articles are selected by topic modeling, including six impact categories (infrastructural, political, financial, ecological, agricultural, and human health), which are informed by previous studies (Imran et al., 2016a) and align with modern disaster impact assessment frameworks (Silva et al., 2022)." (p.2) |
| Q25 | "Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)                                                                                                                        |
| Q28 | "Unlike previous study (Imran et al., 2016a), however, each sample might correspond to more than one impact." (p.4)                                                                                                                                                                                                      |
| D10 | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                    |

## Checklist

**Checklist item definitions:**

| ID   | Assessment Question                                            |
|------|----------------------------------------------------------------|
| IO-1 | Identify test case categories required for regional deployment |
| IO-2 | Check if taxonomy omits regionally relevant categories         |
| IO-3 | Check if taxonomy includes irrelevant categories               |
| IO-4 | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** Required regional categories include riverine flooding with cross-border displacement, locust outbreaks, Chaco fires, dirt-road access disruption to indigenous communities, agrarian supply chain interruption, and border integrity concerns [WEB-5, WEB-1, WEB-2].

**IO-2:** Yes — the regional YAML explicitly identifies locust outbreaks and cross-border displacement as ABSENT from the benchmark's 15 weather event types and impact category definitions. Dataset analysis confirms no Southern Cone or border-integrity exemplars in 31 sampled examples [D10, D18, D19].

**IO-3:** The 15 LDA-derived weather event types are Canadian-newspaper-derived [Q97, Q101] and include urban North American snowstorm/blizzard event types (e.g., 1894 blizzard [D3], Quebec ice storm [D9], Montreal municipal snow removal [D10]) that are not weather event types relevant to the semi-arid Gran Chaco context.

**IO-4:** Critical content validity gaps: (a) no cross-border displacement category, (b) no locust outbreak event type, (c) no Chaco fire event type, (d) no indigenous-community-access damage type, (e) no consequence-framing mechanism for ambiguous multi-label assignment as practiced by Gran Chaco coordinators.

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q11   | "The articles are selected by topic modeling, including six impact categories (infrastructural, political, financial, ecological, agricultural, and human health), which are informed by previous studies (Imran et al., 2016a) and align with modern disaster impact assessment frameworks (Silva et al., 2022)." (p.2)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Q97   | "Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Q101  | "Using Latent Dirichlet Allocation, the dataset was categorized into 15 primary weather event types." (p.14)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Q140  | "Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| D3    | by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| D9    | residents of southwestern Quebec have been plunged back into the deep freeze Overnight lows in Montreal were expected to drop to minus-16 Celsius last night and tumble as far as minus-30C on the South Shore, where more than one million people struggle on without power or heat — 1998 Quebec ice storm aftermath — one million without power/heat, annotated all-zero                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| D10   | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| D18   | A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — 1992 Mediterranean flash flood — political and ecological but not infrastructural ("swept away cars," "uprooted trees") — label boundary question                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| D19   | With about half the original 1.6-million-hectare swamp filled for development or drained for agriculture, the park includes about 202,000 hectares of marsh experts liken the Everglades to a seriously ill patient — 1990 Everglades drought/ecosystem article — ecological only; US content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| WEB-1 | Paraguayan emergency Ley N° 7471/2025 frames flood emergency at the departmental level — political framing absent from benchmark exemplars ( <a href="https://www.bacn.gov.py/leyes-paraguayas/12739/ley-n-7471-2025-que-declara-en-situaci-n-de-emergenci-a-a-los-departamentos-de-presidente-hayes-boquer-n-y-alto-paraguay-y-ampl-a-el-presupuesto-general-de-la-naci-n-para-el-ejerci-cio-fiscal-2025-aprobado-por-ley-n-7408-de-fecha-30-de-diciembre-de-2024-presidencia-de-la-rep-blica-secretar-a-de-emergencia-nacional">https://www.bacn.gov.py/leyes-paraguayas/12739/ley-n-7471-2025-que-declara-en-situaci-n-de-emergenci-a-a-los-departamentos-de-presidente-hayes-boquer-n-y-alto-paraguay-y-ampl-a-el-presupuesto-general-de-la-naci-n-para-el-ejerci-cio-fiscal-2025-aprobado-por-ley-n-7408-de-fecha-30-de-diciembre-de-2024-presidencia-de-la-rep-blica-secretar-a-de-emergencia-nacional</a> ) |
| WEB-2 | Route No. 12 along the Pilcomayo is impassable when raining — primary infrastructural concern absent from benchmark ( <a href="https://en.wikipedia.org/wiki/Presidente_Hayes_Department">https://en.wikipedia.org/wiki/Presidente_Hayes_Department</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|              |                                                                                                                                                                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-5</b> | Cross-border displacement and indigenous community evacuation are primary Gran Chaco disaster consequences (10,000+ Wichi/Toba/Chorote/Tapiete evacuated in 2018) ( <a href="https://www.iadb.org/en/staying-ahead-gran-chacos-floods">https://www.iadb.org/en/staying-ahead-gran-chacos-floods</a> ) |
| <b>WEB-6</b> | Pilcomayo basin spans ~50,000 km <sup>2</sup> with cross-border community coordination needs ( <a href="https://www.wri.org/insights/gran-chaco-communities-build-climate-resilience">https://www.wri.org/insights/gran-chaco-communities-build-climate-resilience</a> )                              |

## Information Gaps

- No publicly available Spanish-language disaster impact NLP benchmark exists for any Southern Cone context against which to compare wximpactbench's gaps quantitatively

## Requires Expert Verification

- AFE/SINAGIR national coordinators should confirm whether the six categories, even with definitional expansion, are operationally adequate or whether additional categories (e.g., a dedicated Border Integrity category) are required

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## Remediation

**Gap 1: Six categories and 15 weather event types are Canadian-derived; cross-border displacement, locust outbreaks, Chaco fires, and indigenous-community-access disruption are absent.**

**Recommendation:** Convene an AFE/SINAGIR–SEN–VIDECI working group to extend the taxonomy with at minimum: a Border Integrity / Cross-Border Displacement category; locust outbreak and Chaco fire weather event types; an indigenous-community-access subdimension within Infrastructural and Human Health categories.

# Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Gran Chaco Tri-Border Emergency Response Zone

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark's entire input corpus is English-language formal newspaper prose from three Canadian newspapers (La Presse, La Patrie, Montreal Gazette) accessed via McGill University Library and the Bibliothèque nationale du Québec [Q97]. Dataset analysis confirmed every one of 31 sampled examples is English Canadian newspaper content; no Spanish, no social media, no indigenous-language toponyms, no Southern Cone references. The deployment requires Rio Platense Spanish social media and local news with Guaraní/Quechua/Wichí lexical borrowing [WEB-7, WEB-9, WEB-10]. The benchmark explicitly acknowledges potential biases in 'underrepresented historical events and linguistic variations' [Q90] but does not extend to Spanish-language or Southern Cone content. The regional context confirms no Spanish-language disaster NLP benchmark exists for this region.

| ID     | Evidence                                                                                                                                                                                                                                                                                  |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q90    | "Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations." (p.9) |
| Q97    | "Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec." (p.9)                                           |
| WEB-7  | Rio Platense Spanish is the predominant variety in Argentina/Uruguay extending to Paraguay and border regions ( <a href="https://en.wikipedia.org/wiki/Rioplatense_Spanish">https://en.wikipedia.org/wiki/Rioplatense_Spanish</a> )                                                       |
| WEB-9  | No NLP tools handle indigenous-language lexical borrowing within Spanish disaster text ( <a href="https://arxiv.org/html/2501.09943v2">https://arxiv.org/html/2501.09943v2</a> )                                                                                                          |
| WEB-10 | No mixed-language disaster classification tools exist for Guaraní/Quechua/Wichí lexical borrowing within Spanish ( <a href="https://arxiv.org/html/2404.05365v1">https://arxiv.org/html/2404.05365v1</a> )                                                                                |

## Strengths

- Two temporal periods (historical 1880s–1890s; modern 1990s–2000s) provide some intra-corpus register variation [Q13, D3, D16]
- Some non-Canadian wire-service content (China, Indonesia, US, Europe) appears in the labeled data [D7, D8, D16, D19], showing the corpus is not strictly Canadian — though still not Southern Cone

| ID  | Evidence                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q13 | "We extract a sample of articles from two time periods, which cover linguistic and social changes across different eras and increase linguistic complexity due to the different descriptions of weather events in different times (Campbell, 2013)." (p.2)                                                                                                                  |
| D3  | by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample                                                                                                                                         |
| D7  | Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged — 1996 Yangtze River flood report from China — not Canadian, annotated for infrastructure                                                                                                                                       |
| D8  | The Jakarta Post reported that hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries — Indonesian volcanic eruption and Washington state earthquake — non-Canadian, non-weather-disaster content labeled for infrastructure and health                           |
| D16 | More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged |

|            |                                                                                                                                                                                                                                                                                               |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D19</b> | With about half the original 1.6-million-hectare swamp filled for development or drained for agriculture, the park includes about 202,000 hectares of marsh experts liken the Everglades to a seriously ill patient — 1990 Everglades drought/ecosystem article — ecological only; US content |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                                        |
|-------------|----------------------------------------------------------------------------|
| <b>IC-1</b> | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| <b>IC-2</b> | Does culturally sensitive content align with target culture?               |
| <b>IC-3</b> | Flag inputs assuming Western-specific knowledge                            |
| <b>IC-4</b> | Regional annotator recruitment for culturally sensitive instances          |
| <b>IC-5</b> | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — deployment inputs require knowledge of Rio Platense Spanish dialectal features, Guaraní/Quechua/Wichí toponyms (Pilcomayo, Bermejo, Tartagal, Yacuiba), and Paraguayan Jopara mixed register [WEB-7, WEB-9]; benchmark provides none of this [D1–D22 all English].

**IC-2:** No — benchmark content reflects North American journalistic culture (e.g., Montreal municipal snow removal [D10], US Everglades policy [D19]) rather than Southern Cone disaster-response culture [Q97].

**IC-3:** Yes — content frequently requires understanding of Canadian/US institutional context (Toronto electric car services [D3], Quebec ice storm [D9], US Supreme Court [D22], Berlin Wall anniversary [D5]) that does not transfer to Gran Chaco deployment.

**IC-4:** Regional annotator recruitment has not occurred; no AFE/SINAGIR, SEN, VIDECI, or Gendarmería personnel were involved in benchmark construction [Q110].

**IC-5:** Critical content validity violation: zero linguistic, geographic, or cultural overlap between benchmark inputs and deployment input distribution; results from this benchmark cannot be transferred to predict performance on Gran Chaco Spanish social media.

| ID          | Evidence                                                                                                                                                                                                                                                                                                |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q90</b>  | “Although WXIMPACTBENCH provides valuable insights (e.g., exhibit the strengths and weaknesses of various society impact understanding) about evaluating LLMs on disruptive weather, it may have potential biases in underrepresented historical events and linguistic variations.” (p.9)               |
| <b>Q97</b>  | “Our primary data source is a corpus of three digitized newspapers (La Presse, La Patrie and Montreal Gazette), obtained through collaboration with the McGill University Library and Archives and the Bibliothèque nationale du Québec.” (p.9)                                                         |
| <b>Q110</b> | “The annotation process was conducted by members of a research group specializing in uncovering the history of a region’s climate change through the regional historical weather records.” (p.15)                                                                                                       |
| <b>D1</b>   | John’s Cloudy -4 -4 United States Max Min Atlanta Cloudy 17 4 Boston Cloudy 3 -6 Chicago Cloudy 2 2 Dallas Thunderstorms 11 2 — Weather table/tabular data with no narrative — all-zero label, non-event content                                                                                        |
| <b>D2</b>   | But the real beneficiaries of the work-at-home trend are the employees themselves... She takes a turn at the table in a lay way (5) 29, A good one is of assistance (9) — Crossword puzzle clues and work-from-home editorial — zero weather impact content                                             |
| <b>D3</b>   | by six o’clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample                                                                     |
| <b>D4</b>   | Police say the aggravated assault charge against Clarkson involves a February attack on Boland in which the celebrity photographer’s face and chest were splashed with concentrated sulphuric acid. — Crime report with no weather content — all-zero label                                             |
| <b>D5</b>   | The decision yesterday followed a 16-hour overnight meeting in Berlin, where debate over how much to give away was sometimes interrupted by the sounds of revellers celebrating the first anniversary of the opening of the Berlin Wall. — Political news about German unification — no weather content |



|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D6</b>     | A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — Storm narrative with human casualty but annotated all-zero — potential label discrepancy                                                                                                                                                               |
| <b>D7</b>     | Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged — 1996 Yangtze River flood report from China — not Canadian, annotated for infrastructure                                                                                                                                                                                                             |
| <b>D8</b>     | The Jakarta Post reported that hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries — Indonesian volcanic eruption and Washington state earthquake — non-Canadian, non-weather-disaster content labeled for infrastructure and health                                                                                                 |
| <b>D9</b>     | residents of southwestern Quebec have been plunged back into the deep freeze Overnight lows in Montreal were expected to drop to minus-16 Celsius last night and tumble as far as minus-30C on the South Shore, where more than one million people struggle on without power or heat — 1998 Quebec ice storm aftermath — one million without power/heat, annotated all-zero                                                                       |
| <b>D10</b>    | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                                                                                                                                             |
| <b>D11</b>    | -2 City res y 5000 5 5 5 Aieaxavn 8000 50 50 50 Ckearcdnf 835 825 25 JS 1 Almaden 1000 35 35 35 — Stock ticker/financial table — all-zero label, clearly off-topic content                                                                                                                                                                                                                                                                        |
| <b>D12</b>    | The walkout will interrupt regular garbage collection and will affect street and sidewalk cleanup if it snows. Unless nine centimetres or more of snow falls during the walkout, blue-collar workers aren't obliged to clean the streets and sidewalks. — Labor dispute/garbage collection strike — weather mentioned incidentally, all-zero label                                                                                                |
| <b>D13</b>    | Penn refuses to play up the easy nebbish... he does not wear glasses, flood pants or a bow tie — Film review with only figurative use of "flood" — all-zero label, entirely off-topic                                                                                                                                                                                                                                                             |
| <b>D14</b>    | in Richmond, the storm so long expected has broken forth in all its fury. Great flashes of blinding lightning intermingled with the grand roar of the thunder from on high — Fictional storm description used metaphorically in a society/romance narrative — all-zero label                                                                                                                                                                      |
| <b>D15</b>    | the agency said Christian Lambrechts... said this would expose more of the sea's surface to sunlight, rather than reflect it, contributing to continued and accelerated warming The peninsula - the tongue of land that juts toward South America - has been hit by greater warming than almost any other area on Earth — Antarctic/climate change science brief — no direct disaster impact, all-zero label                                      |
| <b>D16</b>    | More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged                                                                       |
| <b>D17</b>    | 01 SECOND RACE: Trot, I Mile Purse: 11,000 5 Prime Saliora (M. Lalonde) 9.70 5.10 5.70 3 Bidou Fio (R. SimarrJ) 5.30 4.40 — Horse racing results table with unrelated editorial on optometrists' ad — off-topic noise                                                                                                                                                                                                                             |
| <b>D18</b>    | A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — 1992 Mediterranean flash flood — political and ecological but not infrastructural ("swept away cars," "uprooted trees") — label boundary question |
| <b>D19</b>    | With about half the original 1.6-million-hectare swamp filled for development or drained for agriculture, the park includes about 202,000 hectares of marsh experts liken the Everglades to a seriously ill patient — 1990 Everglades drought/ecosystem article — ecological only; US content                                                                                                                                                     |
| <b>D20</b>    | July 7th they encountered a heavy gale from the west, and had great difficulty in keeping the boat free, the sea continually breaking on board... After the accident they suffered severely from the cold. — 1896 Atlantic rowing voyage encounter with gale — human suffering described but labeled all-zero                                                                                                                                     |
| <b>D21</b>    | TRAFFIC IS PARALYZED In Western Canadian Cities, and at Many Points In the United States Disasters In England. — 1894 blizzard chunk (segmented from Ex. 3) — infrastructure label carried over to chunk                                                                                                                                                                                                                                          |
| <b>D22</b>    | Peter Russell, a law professor at the University of Toronto, said the Supreme Court will benefit from Binnie's solid experience in constitutional and international affairs. — Legal/judicial appointment news — entirely off-topic                                                                                                                                                                                                               |
| <b>WEB-5</b>  | Cross-border displacement and indigenous community evacuation are primary Gran Chaco disaster consequences (10,000+ Wichi/Toba/Chorote/Tapiete evacuated in 2018) ( <a href="https://www.iadb.org/en/staying-ahead-gran-chacos-floods">https://www.iadb.org/en/staying-ahead-gran-chacos-floods</a> )                                                                                                                                             |
| <b>WEB-7</b>  | Rio Platense Spanish is the predominant variety in Argentina/Uruguay extending to Paraguay and border regions ( <a href="https://en.wikipedia.org/wiki/Rioplatense_Spanish">https://en.wikipedia.org/wiki/Rioplatense_Spanish</a> )                                                                                                                                                                                                               |
| <b>WEB-9</b>  | No NLP tools handle indigenous-language lexical borrowing within Spanish disaster text ( <a href="https://arxiv.org/html/2501.09943v2">https://arxiv.org/html/2501.09943v2</a> )                                                                                                                                                                                                                                                                  |
| <b>WEB-11</b> | RoBERTuito (500M+ Spanish tweets) is the most relevant existing model for the deployment's input register, but is not disaster-domain-specific and not regionally evaluated ( <a href="https://aclanthology.org/2022.lrec-1.785/">https://aclanthology.org/2022.lrec-1.785/</a> )                                                                                                                                                                 |

## Information Gaps



- No quantitative characterization of actual Gran Chaco disaster social media token-length and lexical-borrowing distribution exists; would require corpus collection from regional platforms

## Requires Expert Verification

- Regional annotators (AFE/SINAGIR coordinators, SEN, VIDECl personnel) should validate which proportion of routine deployment inputs contain indigenous-language lexical borrowing requiring specialized handling
- 

## Remediation

**Gap 1: Zero Spanish-language, social-media, or Southern Cone content; no indigenous-language lexical-borrowing exemplars.**

**Recommendation:** Construct a Gran Chaco Spanish-language evaluation corpus from Twitter/X, Facebook community groups, WhatsApp broadcast channels, and local news archives spanning Salta/Formosa/Chaco/Boquerón/Alto Paraguay/Presidente Hayes/Tarija; include an indigenous-toponym/lexical-borrowing test slice.

# Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Gran Chaco Tri-Border Emergency Response Zone

**Input Form definition:** Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Benchmark inputs are English-language OCR-corrected historical newspaper prose averaging 2,987 tokens (long-context) or 781 tokens (mixed-context) [Q109]. Models must support 8k token input lengths [Q55] and handle 'outdated terminology, spelling variations, and evolving writing conventions' [Q14]. Deployment requires processing informal Rio Platense Spanish social media (typically 20–280 tokens per post) with platform abbreviations, hashtags, indigenous-language toponyms, and code-switching [WEB-11, regional YAML register\_variation]. The noise patterns in the benchmark (OCR column-fragmentation artifacts [D11, D17, D21]) are unrelated to the noise patterns in the deployment (informal orthography, abbreviations like 'tmbn', 'q pasa'). RoBERTuito-class social-media-tuned Spanish models exist [WEB-11] but the benchmark provides no evaluation in that direction.

| ID     | Evidence                                                                                                                                                                                                                                                                       |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q14    | "Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013)." (p.2) |
| Q55    | "The used models for the two tasks cover different sizes and support the input length of at least 8k tokens." (p.6)                                                                                                                                                            |
| Q109   | "The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings." (p.15)                                                                                                                                                      |
| D11    | -2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25 JS 1 Almaden 1000 35 35 35 — Stock ticker/financial table — all-zero label, clearly off-topic content                                                                                                      |
| D17    | 01 SECOND RACE: Trot, I Mile Purse: 11,000 5 Prime Saliora (M. Lalonde) 9.70 5.10 5.70 3 Bidou Fio (R. SimarrJ) 5.30 4.40 — Horse racing results table with unrelated editorial on optometrists' ad — off-topic noise                                                          |
| D21    | TRAFFIC IS PARALYZED In Western Canadian Cities, and at Many Points In the United States Disasters In England. — 1894 blizzard chunk (segmented from Ex. 3) — infrastructure label carried over to chunk                                                                       |
| WEB-11 | RoBERTuito (500M+ Spanish tweets) is the most relevant existing model for the deployment's input register, but is not disaster-domain-specific and not regionally evaluated (https://aclanthology.org/2022.lrec-1.785/)                                                        |

## Strengths

- Mixed-context version with ~250-token chunks [Q46] is closer to deployment token lengths than long-context version
- Benchmark explicitly acknowledges register/style variation matters [Q14, Q32], even though only English historical→modern variation is tested
- Models evaluated must support 8k+ tokens [Q55], which is more than sufficient for short Spanish social media posts

| ID  | Evidence                                                                                                                                                                                                                                                                       |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q14 | "Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013)." (p.2) |
| Q32 | "Different from them, our constructed samples require the models to understand the linguistic shifts between historical and modern texts and address inconsistent styles of narratives across various periods." (p.5)                                                          |
| Q46 | "For the long-context impact evaluation, we create an alternate version (mixed context), whose sample length is split into segments with approximately 250 tokens from the original one (long-context version) following (Levy et al., 2024)." (p.6)                           |
| Q55 | "The used models for the two tasks cover different sizes and support the input length of at least 8k tokens." (p.6)                                                                                                                                                            |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                          |
|------|--------------------------------------------------------------|
| IF-1 | Compare signal distributions between source and target       |
| IF-2 | Check if regional infrastructure supports data capture specs |
| IF-3 | Identify domain-specific form differences                    |
| IF-4 | Document form mismatches harming external validity           |

### Checklist responses:

**IF-1:** Significant signal distribution mismatch: English formal newspaper prose vs. Spanish informal social media; OCR-noise vs. social-media-noise; 781–2,987 token articles vs. 20–280 token posts [[Q109](#), [Q120](#), [regional register\\_variation](#)].

**IF-2:** Deployment infrastructure does not produce OCR-style inputs; it ingests live Twitter/X, Facebook, WhatsApp text streams [[WEB-5](#), [regional cultural\\_norms\\_notes](#)]. Connectivity in the Gran Chaco is documented as patchy [[WEB-5](#)], which affects input freshness but not input form per se.

**IF-3:** Domain-specific form differences are severe: benchmark cannot evaluate handling of hashtags, platform abbreviations, voseo, indigenous toponyms, or Jopara code-mixing [[WEB-9](#), [WEB-10](#)].

**IF-4:** Critical external validity violation: model performance on benchmark's English long-form OCR-corrected prose cannot predict performance on Spanish short-form informal social media posts.

| ID                     | Evidence                                                                                                                                                                                                                                                                                              |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q14</a>    | "Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013)." (p.2)                        |
| <a href="#">Q32</a>    | "Different from them, our constructed samples require the models to understand the linguistic shifts between historical and modern texts and address inconsistent styles of narratives across various periods." (p.5)                                                                                 |
| <a href="#">Q109</a>   | "The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings." (p.15)                                                                                                                                                                             |
| <a href="#">Q120</a>   | "Figure 6 presents the token length distribution of passages in two versions of our dataset: (a) the Long Context dataset and (b) the Mixed Context dataset used for context-denoising evaluation." (p.16)                                                                                            |
| <a href="#">D3</a>     | by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample                                                                   |
| <a href="#">D11</a>    | -2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25 JS 1 Almaden 1000 35 35 35 — Stock ticker/financial table — all-zero label, clearly off-topic content                                                                                                                             |
| <a href="#">D21</a>    | TRAFFIC IS PARALYZED In Western Canadian Cities, and at Many Points In the United States Disasters In England. — 1894 blizzard chunk (segmented from Ex. 3) — infrastructure label carried over to chunk                                                                                              |
| <a href="#">WEB-5</a>  | Cross-border displacement and indigenous community evacuation are primary Gran Chaco disaster consequences (10,000+ Wichi/Toba/Chorote/Tapiete evacuated in 2018) ( <a href="https://www.iadb.org/en/staying-ahead-gran-chacos-floods">https://www.iadb.org/en/staying-ahead-gran-chacos-floods</a> ) |
| <a href="#">WEB-9</a>  | No NLP tools handle indigenous-language lexical borrowing within Spanish disaster text ( <a href="https://arxiv.org/html/2501.09943v2">https://arxiv.org/html/2501.09943v2</a> )                                                                                                                      |
| <a href="#">WEB-10</a> | No mixed-language disaster classification tools exist for Guaraní/Quechua/Wichí lexical borrowing within Spanish ( <a href="https://arxiv.org/html/2404.05365v1">https://arxiv.org/html/2404.05365v1</a> )                                                                                            |
| <a href="#">WEB-11</a> | RoBERTuito (500M+ Spanish tweets) is the most relevant existing model for the deployment's input register, but is not disaster-domain-specific and not regionally evaluated ( <a href="https://aclanthology.org/2022.lrec-1.785/">https://aclanthology.org/2022.lrec-1.785/</a> )                     |

## Information Gaps

- Empirical token-length and noise-pattern distribution for Gran Chaco disaster social media has not been characterized

## Requires Expert Verification

- A small Spanish-social-media replication study (e.g., evaluating RoBERTuito on Gran Chaco posts) would directly establish form-mismatch impact
- 

## Remediation

**Gap 1: Benchmark form is English long-form OCR-corrected prose; deployment form is short Spanish social media with informal orthography and code-mixing.**

**Recommendation:** Pilot RoBERTuito and multilingual LLMs on a Spanish social-media disaster classification task, using token-length and noise-pattern distributions matching deployment inputs; report results separately from wximpactbench scores.

# Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Gran Chaco Tri-Border Emergency Response Zone

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark uses six binary impact labels [Q25, Q117] with detailed Canadian-anchored operational definitions [Q141–Q146] and a multi-label simultaneous classification structure [Q57] that nominally matches the deployment's minimum acceptable output form. However, the boundary logic is misaligned with the consequence-framing-dependent assignment that Gran Chaco national coordinators apply. Dataset analysis reveals visible label boundary inconsistencies even within the Canadian context: 100,000 homes damaged labeled agricultural+health but not infrastructural [D16]; storm with skull-fracture casualty labeled all-zero [D6]; one million people without power labeled all-zero [D9]. The Political category exemplars (municipal snow removal [D10], policy debates [Q140]) lack any border integrity or cross-border displacement framing. The Agricultural category emphasizes Canadian farm-context exemplars [Q136, Q142] without dirt-road access disruption to indigenous communities. Score is 2 (not 1) because the multi-label binary structure itself is reusable and partially aligned with deployment needs.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q25  | "Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)                                                                                                                                                                                                                                                                                                                                                                                              |
| Q57  | "Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6)                                                                                                                                                                                                                                                                                                       |
| Q117 | "The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Q136 | "Agricultural Impact: Focuses on weather-related effects on farming and livestock management. Includes crop yield variations; direct damage to crops, timber, or livestock; modifications to farming schedules; disruptions to food production and supply chains; impacts on farming equipment; and changes in agricultural inputs (e.g., soil conditions, water availability, fertilizers, animal feed). Both immediate and long-term effects should be considered." (p.19)                                                                                                                   |
| Q140 | "Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)                                                                                                                                                                                             |
| Q141 | "Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification." (p.20)                              |
| Q142 | "Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed." (p.20) |
| Q143 | "Ecological Impact: Classify as 'true' if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life." (p.20)                                                                                             |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q144</b> | “Financial Impact: Classify as ‘true’ if the text explicitly mentions economic consequences of weather events. This includes direct monetary losses; business disruptions or closures requiring financial intervention; market price fluctuations or demand changes for specific goods; impacts on tourism and local economic activities; or insurance claims or economic relief measures. Focus only on explicit mentions of financial losses or fluctuations.” (p.20) |
| <b>Q145</b> | “Human Health Impact: Classify as ‘true’ if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications.” (p.20)          |
| <b>Q146</b> | “Political Impact: Classify as ‘true’ if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities.” (p.20)                                       |
| <b>D6</b>   | A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — Storm narrative with human casualty but annotated all-zero — potential label discrepancy                                                                                                                                                                                     |
| <b>D9</b>   | residents of southwestern Quebec have been plunged back into the deep freeze Overnight lows in Montreal were expected to drop to minus-16 Celsius last night and tumble as far as minus-30C on the South Shore, where more than one million people struggle on without power or heat — 1998 Quebec ice storm aftermath — one million without power/heat, annotated all-zero                                                                                             |
| <b>D10</b>  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                                                                                                                                                                   |
| <b>D16</b>  | More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged                                                                                             |

## Strengths

- Multi-label simultaneous binary structure [Q57, Q117] matches the deployment's minimum acceptable output form
- Each sample may carry multiple labels simultaneously [Q28, D10] — structurally compatible with consequence-framing where one event attracts multiple labels
- Detailed operational definitions exist [Q135–Q146], providing a starting template that could be adapted/expanded for Gran Chaco contexts

| ID          | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q28</b>  | “Unlike previous study (Imran et al., 2016a), however, each sample might correspond to more than one impact.” (p.4)                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Q57</b>  | “Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once.” (p.6)                                                                                                                                                                                            |
| <b>Q117</b> | “The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts.” (p.16)                                                                                                                                                                                                                                                                                                                           |
| <b>Q135</b> | “Infrastructural Impact: Examines weather-related damage or disruption to physical infrastructure and essential services. Includes structural damage to buildings, roads, and bridges; disruptions to transportation (e.g., railway cancellations, road closures); interruptions to utilities (e.g., power, water supply); failures in communication networks; and industrial facility damage. Both immediate physical damage and service disruptions should be considered.” (p.19) |
| <b>Q136</b> | “Agricultural Impact: Focuses on weather-related effects on farming and livestock management. Includes crop yield variations; direct damage to crops, timber, or livestock; modifications to farming schedules; disruptions to food production and supply chains; impacts on farming equipment; and changes in agricultural inputs (e.g., soil conditions, water availability, fertilizers, animal feed). Both immediate and long-term effects should be considered.” (p.19)        |
| <b>Q137</b> | “Ecological Impact: Examines effects on natural environments and ecosystems. Includes changes in biodiversity; impacts on wildlife populations and behavior; effects on non-agricultural plant life; habitat modifications (e.g., forests, wetlands, water bodies); changes in hydrological systems (e.g., river levels, lake conditions); and urban plant life impact. Immediate environmental changes should be prioritized.” (p.19)                                              |
| <b>Q138</b> | “Financial Impact: Analyzes economic consequences of weather events. Includes direct monetary losses; business disruptions requiring financial intervention; market fluctuations; impacts on tourism and local economies; and insurance claims or economic relief measures. The focus should be on explicit financial impacts rather than inferred consequences.” (p.19)                                                                                                            |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q139</b> | "Human Health Impact: Examines both physical and mental health effects. Includes direct injuries or fatalities (including cases where one or more casualties are explicitly mentioned); increased risks of weather-related illnesses; mental health consequences (e.g., stress, anxiety); impacts on healthcare accessibility; and long-term health implications. Both short-term and chronic health effects should be considered." (p.19)                                                                                                                                                     |
| <b>Q140</b> | "Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)                                                                                                                                                                                             |
| <b>Q141</b> | "Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification." (p.20)                              |
| <b>Q142</b> | "Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed." (p.20) |
| <b>Q143</b> | "Ecological Impact: Classify as 'true' if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life." (p.20)                                                                                             |
| <b>Q144</b> | "Financial Impact: Classify as 'true' if the text explicitly mentions economic consequences of weather events. This includes direct monetary losses; business disruptions or closures requiring financial intervention; market price fluctuations or demand changes for specific goods; impacts on tourism and local economic activities; or insurance claims or economic relief measures. Focus only on explicit mentions of financial losses or fluctuations." (p.20)                                                                                                                        |
| <b>Q145</b> | "Human Health Impact: Classify as 'true' if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications." (p.20)                                                                                                                                 |
| <b>Q146</b> | "Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)                                                                                                                                                              |
| <b>D10</b>  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                                                                                                                                                                                                                                                                                          |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                          |
|-------------|--------------------------------------------------------------|
| <b>OO-1</b> | Review output label categories for regional relevance        |
| <b>OO-2</b> | Identify missing regionally specific categories              |
| <b>OO-3</b> | Flag categories encoding non-regional values                 |
| <b>OO-4</b> | Consider stakeholder-driven taxonomy redesign                |
| <b>OO-5</b> | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** Six labels nominally cover broad disaster consequences but operationalize them via Canadian exemplars [\[Q141–Q146\]](#); boundary logic does not match Gran Chaco consequence-framing [\[regional\\_consequence\\_framing\\_dependency\]](#).

**OO-2:** Missing categories: cross-border displacement (regional-priority, ABSENT), border integrity (ABSENT), indigenous-community-access disruption, strategic economic route disruption, locust outbreaks. Confirmed by [\[WEB-5, WEB-1\]](#) and dataset gap [\[D10, D18\]](#).



**OO-3:** Political category encodes North American policy/governance assumptions (municipal budget debates, electoral processes) [Q140, D10]; Ecological category centers North American ecosystems (Everglades [D19]); neither encodes Southern Cone or border-region values.

**OO-4:** Stakeholder-driven taxonomy redesign would be required: AFE/SINAGIR, SEN, and VIDEI national coordinators would need to define operational labels reflecting consequence-framing logic [WEB-16, WEB-20, WEB-22].

**OO-5:** Structural validity is compromised by the operational definitions; content validity is compromised by missing categories; external validity is compromised by definitional drift between Canadian and Gran Chaco framings.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q25  | "Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)                                                                                                                                                                                                                                                                                                                                                                                              |
| Q57  | "Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6)                                                                                                                                                                                                                                                                                                       |
| Q117 | "The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Q140 | "Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)                                                                                                                                                                                             |
| Q141 | "Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification." (p.20)                              |
| Q142 | "Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed." (p.20) |
| Q143 | "Ecological Impact: Classify as 'true' if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life." (p.20)                                                                                             |
| Q144 | "Financial Impact: Classify as 'true' if the text explicitly mentions economic consequences of weather events. This includes direct monetary losses; business disruptions or closures requiring financial intervention; market price fluctuations or demand changes for specific goods; impacts on tourism and local economic activities; or insurance claims or economic relief measures. Focus only on explicit mentions of financial losses or fluctuations." (p.20)                                                                                                                        |
| Q145 | "Human Health Impact: Classify as 'true' if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications." (p.20)                                                                                                                                 |
| Q146 | "Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)                                                                                                                                                              |
| Q147 | "Return 'false' for any impact category that is either not present in the text or not related to weather events. Base classifications on explicit mentions in the text. Focus on direct impacts rather than implications. Consider immediate and direct effects." (p.20)                                                                                                                                                                                                                                                                                                                       |
| D10  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                                                                                                                                                                                                                                                                                                                          |
| D16  | More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged                                                                                                                                                                                                                    |
| D18  | A line of freak thunderstorms rumbling across the northeastern Mediterranean coast from Spain and southern France to the principality of Monaco brought two days of disastrous flash flooding... Raging waters uprooted trees and swept away cars as river levels rose by as much as 3 metres — 1992 Mediterranean flash flood — political and ecological but not infrastructural ("swept away cars," "uprooted trees") — label boundary question                                                                                                                                              |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D19    | With about half the original 1.6-million-hectare swamp filled for development or drained for agriculture, the park includes about 202,000 hectares of marsh experts liken the Everglades to a seriously ill patient — 1990 Everglades drought/ecosystem article — ecological only; US content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| WEB-1  | Paraguayan emergency Ley N° 7471/2025 frames flood emergency at the departmental level — political framing absent from benchmark exemplars ( <a href="https://www.bacn.gov.py/leyes-paraguayas/12739/ley-n-7471-2025-que-declara-en-situacion-de-emergencia-a-los-departamentos-de-presidente-hayes-boquer-n-y-alto-paraguay-y-ampl-a-el-presupuesto-general-de-la-nacion-para-el-ejercicio-fiscal-2025-aprobado-por-ley-n-7408-de-fecha-30-de-diciembre-de-2024-presidencia-de-la-republica-secretaria-de-emergencia-nacional">https://www.bacn.gov.py/leyes-paraguayas/12739/ley-n-7471-2025-que-declara-en-situacion-de-emergencia-a-los-departamentos-de-presidente-hayes-boquer-n-y-alto-paraguay-y-ampl-a-el-presupuesto-general-de-la-nacion-para-el-ejercicio-fiscal-2025-aprobado-por-ley-n-7408-de-fecha-30-de-diciembre-de-2024-presidencia-de-la-republica-secretaria-de-emergencia-nacional</a> ) |
| WEB-5  | Cross-border displacement and indigenous community evacuation are primary Gran Chaco disaster consequences (10,000+ Wichí/Toba/Chorote/Tapiete evacuated in 2018) ( <a href="https://www.iadb.org/en/staying-ahead-gran-chacos-floods">https://www.iadb.org/en/staying-ahead-gran-chacos-floods</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| WEB-16 | AFE/SINAGIR is the current Argentine national authority — designated as authoritative labelers by deployment user, with zero benchmark representation ( <a href="https://www.argentina.gob.ar/sinagir/institucional">https://www.argentina.gob.ar/sinagir/institucional</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| WEB-20 | SEN (Paraguay) coordinates emergencies in all three Chaco departments — also unrepresented ( <a href="https://sen.gov.py/institucional/">https://sen.gov.py/institucional/</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| WEB-22 | VIDECI (Bolivia) operates departmental representation in Tarija — also unrepresented ( <a href="https://www.mindef.gob.bo/node/499">https://www.mindef.gob.bo/node/499</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Information Gaps

- Gran Chaco consequence-framing logic has not been formally documented as an annotation guideline

## Requires Expert Verification

- AFE/SINAGIR national coordinators should review the six-category definitions and propose extensions or replacements

## Remediation

**Gap 1: Label boundaries operationalized via Canadian exemplars; consequence-framing-dependent assignment is unsupported.**

**Recommendation:** Rewrite operational definitions ([Q141–Q146]) with Gran Chaco exemplars elicited from AFE/SINAGIR/SEN/VIDECI coordinators; explicitly document consequence-framing decision rules for ambiguous multi-label cases (e.g., flooded dirt road blocking access to indigenous community).

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q141 | “Infrastructure Impact: Classify as ‘true’ if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification.” (p.20)                               |
| Q142 | “Agricultural Impact: Classify as ‘true’ if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed.” (p.20) |
| Q143 | “Ecological Impact: Classify as ‘true’ if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life.” (p.20)                                                                                             |
| Q144 | “Financial Impact: Classify as ‘true’ if the text explicitly mentions economic consequences of weather events. This includes direct monetary losses; business disruptions or closures requiring financial intervention; market price fluctuations or demand changes for specific goods; impacts on tourism and local economic activities; or insurance claims or economic relief measures. Focus only on explicit mentions of financial losses or fluctuations.” (p.20)                                                                                                                        |
| Q145 | “Human Health Impact: Classify as ‘true’ if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications.” (p.20)                                                                                                                                 |

**Q146**

“Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities.” (p.20)

# Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: WXIMPACTBENCH | Context: Gran Chaco Tri-Border Emergency Response Zone

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Annotation was conducted by three Canadian academic researchers from a research group specializing in 'uncovering the history of a region's climate change through the regional historical weather records' [Q110, Q26], reviewed by meteorological experts [Q104]. No inter-annotator agreement statistics are reported. The deployment user designated AFE/SINAGIR national coordinators in Buenos Aires as the authoritative ground-truth labelers — a population with zero representation in the annotation process. The guidelines emphasize 'direct and immediate effects and explicit references' [Q118], an annotation philosophy that produced visible label-correctness concerns in the dataset: explicit weather-caused human casualties labeled all-zero [D6, D20], one million people without power/heat labeled all-zero [D9]. Under Gran Chaco consequence-framing logic [regional\_consequence\_framing\_dependency], these would attract Human Health and Infrastructural labels. The benchmark itself acknowledges interpretation 'might be influenced by demographic and contextual factors' [Q96]. Convergent and external validity are severely compromised.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                    |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q26  | "Annotation is conducted by three domain annotators following our guidelines (provided in Appendix B.2.2)." (p.4)                                                                                                                                                                                                                                                           |
| Q96  | "The interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors, which is similar to other text datasets generated through crowd-sourcing with inherent challenges in ensuring that dataset labels are fully representative of diverse societal perspectives (Talat et al., 2022)." (p.9)             |
| Q104 | "To ensure high-quality and consistent annotations, the task was conducted using a set of specific instructions reviewed by meteorological experts." (p.15)                                                                                                                                                                                                                 |
| Q110 | "The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records." (p.15)                                                                                                                                                                           |
| Q118 | "The guidelines emphasize focusing on direct and immediate effects, ensuring that classifications are based solely on explicit references within the text." (p.16)                                                                                                                                                                                                          |
| D6   | A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — Storm narrative with human casualty but annotated all-zero — potential label discrepancy                                                                                         |
| D9   | residents of southwestern Quebec have been plunged back into the deep freeze Overnight lows in Montreal were expected to drop to minus-16 Celsius last night and tumble as far as minus-30C on the South Shore, where more than one million people struggle on without power or heat — 1998 Quebec ice storm aftermath — one million without power/heat, annotated all-zero |
| D20  | July 7th they encountered a heavy gale from the west, and had great difficulty in keeping the boat free, the sea continually breaking on board... After the accident they suffered severely from the cold. — 1896 Atlantic rowing voyage encounter with gale — human suffering described but labeled all-zero                                                               |

## Strengths

- Annotation guidelines were reviewed by meteorological domain experts [Q104, Q105]
- The same guidelines are embedded in LLM prompts [Q106], creating internal consistency between human and model expectations within the Canadian context
- Mixed-context chunks were independently annotated rather than inheriting parent-article labels [Q47, Q48], which is a methodologically careful choice that produces meaningful negative examples

| ID | Evidence |
|----|----------|
|----|----------|

|             |                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q47</b>  | "Note that annotations for these smaller chunks are performed independently by the same domain experts rather than automatically inherited from the original articles." (p.6)                  |
| <b>Q48</b>  | "This independent annotation process naturally results in some chunks containing no weather impact labels, which serve as valuable negative examples in our evaluation." (p.6)                 |
| <b>Q104</b> | "To ensure high-quality and consistent annotations, the task was conducted using a set of specific instructions reviewed by meteorological experts." (p.15)                                    |
| <b>Q105</b> | "The annotation guideline and the categories definition are provided in Table 14 and Table 15, respectively." (p.15)                                                                           |
| <b>Q106</b> | "Notably, the same instruction guidance is contained within the prompts for LLMs in Appendix C to perform impact classification, following a binary output approach for each category." (p.15) |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                               |
|-------------|-------------------------------------------------------------------|
| <b>OC-1</b> | Do ground-truth labels reflect regional stakeholder perspectives? |
| <b>OC-2</b> | Assess potential annotator-regional population disagreement       |
| <b>OC-3</b> | Review annotator demographics from documentation                  |
| <b>OC-4</b> | Consider label re-annotation by regional annotators               |
| <b>OC-5</b> | Review aggregation methods for minority perspective erasure       |
| <b>OC-6</b> | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** No — labels reflect Canadian academic historical-climate researchers' interpretations [Q110]; AFE/SINAGIR, SEN, and VIDECI emergency coordinators have zero representation [WEB-16, WEB-20, WEB-22].

**OC-2:** Substantial disagreement is empirically demonstrable from dataset analysis: D6 (storm casualty labeled all-zero), D9 (1M without power labeled all-zero), D20 (sailors suffering severely from cold labeled all-zero) — all would likely be re-labeled by Gran Chaco emergency coordinators applying operational salience.

**OC-3:** Annotator demographics: three domain experts from a research group on regional historical climate [Q110, Q26]; no demographic breakdown beyond academic specialization; no datasheet-style documentation.

**OC-4:** Re-annotation by representative AFE/SINAGIR and SEN coordinators on a Spanish-language regional sample is necessary for any deployment-relevant validity claim.

**OC-5:** No inter-annotator agreement reported; aggregation method is not described in detail; minority-perspective erasure is unmeasurable.

**OC-6:** Critical convergent validity (labels do not correlate with regional stakeholder perspectives) and external validity (Canadian academic judgments do not generalize to Gran Chaco operational context) violations.

| ID          | Evidence                                                                                                                                                                                                                                                                                                                                                        |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q26</b>  | "Annotation is conducted by three domain annotators following our guidelines (provided in Appendix B.2.2)." (p.4)                                                                                                                                                                                                                                               |
| <b>Q47</b>  | "Note that annotations for these smaller chunks are performed independently by the same domain experts rather than automatically inherited from the original articles." (p.6)                                                                                                                                                                                   |
| <b>Q96</b>  | "The interpretation of weather-related disruptions in historical newspapers might be influenced by demographic and contextual factors, which is similar to other text datasets generated through crowd-sourcing with inherent challenges in ensuring that dataset labels are fully representative of diverse societal perspectives (Talat et al., 2022)." (p.9) |
| <b>Q110</b> | "The annotation process was conducted by members of a research group specializing in uncovering the history of a region's climate change through the regional historical weather records." (p.15)                                                                                                                                                               |
| <b>Q118</b> | "The guidelines emphasize focusing on direct and immediate effects, ensuring that classifications are based solely on explicit references within the text." (p.16)                                                                                                                                                                                              |

|        |                                                                                                                                                                                                                                                                                                                                                                             |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D6     | A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — Storm narrative with human casualty but annotated all-zero — potential label discrepancy                                                                                         |
| D9     | residents of southwestern Quebec have been plunged back into the deep freeze Overnight lows in Montreal were expected to drop to minus-16 Celsius last night and tumble as far as minus-30C on the South Shore, where more than one million people struggle on without power or heat — 1998 Quebec ice storm aftermath — one million without power/heat, annotated all-zero |
| D16    | More than 292,000 people have been evacuated in the mountainous region along the Yangtze River, with more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed — 2007 European/Chinese/Pakistani multi-location disaster report from Montreal Gazette — agricultural and health labeled, but not infrastructural despite 100k homes damaged |
| D20    | July 7th they encountered a heavy gale from the west, and had great difficulty in keeping the boat free, the sea continually breaking on board... After the accident they suffered severely from the cold. — 1896 Atlantic rowing voyage encounter with gale — human suffering described but labeled all-zero                                                               |
| WEB-16 | AFE/SINAGIR is the current Argentine national authority — designated as authoritative labelers by deployment user, with zero benchmark representation ( <a href="https://www.argentina.gob.ar/sinagir/institucional">https://www.argentina.gob.ar/sinagir/institucional</a> )                                                                                               |
| WEB-20 | SEN (Paraguay) coordinates emergencies in all three Chaco departments — also unrepresented ( <a href="https://sen.gov.py/institucional/">https://sen.gov.py/institucional/</a> )                                                                                                                                                                                            |
| WEB-22 | VIDECI (Bolivia) operates departmental representation in Tarija — also unrepresented ( <a href="https://www.mindef.gob.bo/node/499">https://www.mindef.gob.bo/node/499</a> )                                                                                                                                                                                                |

## Information Gaps

- No inter-annotator agreement statistics; no annotator demographic breakdown beyond research-group specialization

## Requires Expert Verification

- A re-annotation pilot in which AFE/SINAGIR coordinators re-label a sample of benchmark items would empirically establish the convergent-validity gap

## Remediation

**Gap 1: Annotators are Canadian academic climate-history researchers; no AFE/SINAGIR/SEN/VIDECI representation; no IAA reported; visible label-correctness concerns even within Canadian context.**

**Recommendation:** Recruit AFE/SINAGIR national coordinators and SEN/VIDECI departmental representatives as authoritative annotators for a regional re-annotation pilot; report inter-annotator agreement statistics; publish a Datasheet documenting annotator demographics and consequence-framing guidelines.

# Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: WXIMPACTBENCH | Context: Gran Chaco Tri-Border Emergency Response Zone

**Output Form definition:** Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark's multi-label simultaneous binary classification output [Q57, Q117] aligns with the deployment's minimum acceptable output form. Dataset analysis confirms the schema produces compact six-binary label vectors per row [D3, D10], operationally usable for national coordinator dashboards. However, the evaluation framework has notable gaps: F1, accuracy, and row-wise accuracy [Q43, Q45] aggregate across categories and do not differentiate field-gendarme rapid-lookup needs from national-coordinator aggregate-situational-awareness needs. The ranking-based QA task — which the deployment user explicitly rejected as insufficiently reliable for direct operational use — is a substantial portion of the benchmark [Q3, Q15] and includes a known self-referential bias when GPT-4O is both question generator and ranker [Q88]. No confidence scores or probability distributions are produced, although the user indicated these would be useful for human-review-mediated workflows. Score is 3 (mixed) because the classification output structure is genuinely usable, but evaluation-framework relevance to the role-differentiated deployment is partial.

| ID   | Evidence                                                                                                                                                                                                                                                                                 |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q3   | "The benchmark involves two evaluation tasks, multi-label classification and ranking-based question answering." (p.1)                                                                                                                                                                    |
| Q15  | "With our constructed dataset, we develop a benchmark, WXIMPACTBENCH, to investigate the capacity of LLMs to understand disruptive weather impacts with two tasks: i) multi-label classification and ii) ranking-based question-answering." (p.2)                                        |
| Q43  | "For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics." (p.6)                                                                                                                                                                     |
| Q45  | "Compared to the common F1-score and accuracy, the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article." (p.6)                                                                      |
| Q57  | "Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6) |
| Q88  | "Notice that the ranking results would contain bias when the evaluated model is used for question generation (GPT-4O in our cases). This is a common phenomenon (Zhou et al., 2023) and needs to be avoided in benchmarking." (p.9)                                                      |
| Q117 | "The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)                                                                                                                                |
| D3   | by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample                                                      |
| D10  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                    |

## Strengths

- Multi-label simultaneous binary output [Q57] is the minimum acceptable form for the deployment
- Dataset confirms compact binary-vector schema works in practice [D3, D10]
- Standard, well-known classification metrics (F1, accuracy) [Q43] are interpretable by the deployment team
- Modest computational requirements [Q126, Q128] make local replication feasible for deployment teams

| ID  | Evidence                                                                                                             |
|-----|----------------------------------------------------------------------------------------------------------------------|
| Q43 | "For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics." (p.6) |



|             |                                                                                                                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q57</b>  | "Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6) |
| <b>Q126</b> | "For the large proprietary models (e.g., GPT-4o), conducting a one-time evaluation on our WXImpactBench costs approximately \$3 for multi-label classification tasks and \$5.5 for ranking-based QA tasks." (p.17)                                                                       |
| <b>Q128</b> | "The relatively modest computational requirements highlight the accessibility of our benchmark for researchers with limited computational resources, while still enabling comprehensive evaluation of state-of-the-art models" (p.17)                                                    |
| <b>D3</b>   | by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample                                                      |
| <b>D10</b>  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                    |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                    |
|-------------|--------------------------------------------------------|
| <b>OF-1</b> | Does output modality match regional deployment needs?  |
| <b>OF-2</b> | Assess text-to-speech availability for speech output   |
| <b>OF-3</b> | Consider literacy rates and accessibility requirements |
| <b>OF-4</b> | Document form mismatches harming external validity     |

### Checklist responses:

**OF-1:** Output modality (text-based binary multi-label) matches deployment need for simultaneous classification [Q57], but does not provide confidence scores requested for human-review-mediated workflows.

**OF-2:** Not applicable — speech-based output is not a deployment requirement.

**OF-3:** Not directly applicable to text-classification output. Literacy of operators is not in question; the user noted intermediate English proficiency for technical practitioners (system operates in Spanish) [regional\_english\_proficiency].

**OF-4:** Partial external validity gap: F1 and row-wise accuracy [Q43, Q45] do not distinguish field-gendarme rapid-lookup precision needs from national-coordinator aggregate-recall needs; ranking task evaluation is not deployment-relevant per user [regional\_output\_format\_requirements].

| ID          | Evidence                                                                                                                                                                                                                                                                                 |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q43</b>  | "For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics." (p.6)                                                                                                                                                                     |
| <b>Q45</b>  | "Compared to the common F1-score and accuracy, the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article." (p.6)                                                                      |
| <b>Q50</b>  | "For the ranking-based QA task, we deploy the standard metric that emphasizes the accuracy of top positions for evaluation, including Hit@1, nDCG@5, Recall@5, and MRR." (p.6)                                                                                                           |
| <b>Q57</b>  | "Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once." (p.6) |
| <b>Q88</b>  | "Notice that the ranking results would contain bias when the evaluated model is used for question generation (GPT-4O in our cases). This is a common phenomenon (Zhou et al., 2023) and needs to be avoided in benchmarking." (p.9)                                                      |
| <b>Q117</b> | "The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)                                                                                                                                |
| <b>D3</b>   | by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample                                                      |
| <b>D10</b>  | Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already \$30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled                    |

## Information Gaps

- Whether benchmark-style F1 scores predict trust-in-output for field gendarmes vs. national coordinators is unmeasured; no role-differentiated evaluation exists

## Requires Expert Verification

- AFE/SINAGIR coordinators should specify whether per-class precision floors or aggregate recall ceilings are operationally binding metrics
- 

## Remediation

**Gap 1: Aggregate F1/row-wise accuracy metrics do not distinguish field-gendarme precision needs from national-coordinator aggregate-recall needs; no confidence scores produced.**

**Recommendation:** Define role-differentiated evaluation metrics: per-event precision floors for field gendarme rapid-lookup; aggregate per-category recall and calibrated multi-event coverage for national coordinators; expose model probability distributions to support human-in-the-loop review.